NOVA - AI DATA ANALYST

Automated Data Exploration and Predictive Analytics Report

Dataset: Sample Business Sales
Generated on: 04 November 2026, 05:32 AM

1. Dataset Overview

Metric	Value
Number of Rows	106
Number of Columns	10
Total Missing Values	0
Duplicate Rows	0
Numeric Columns	5
Categorical Columns	4

2. Data Quality Comparison

Quality Metric	Before Cleaning	After Cleaning
Rows	107	106
Missing Values	5	0
Duplicate Rows	1	0

3. Data Cleaning Operations

- Removed 1 duplicate rows.
- Converted 'Order_Date' into datetime format.
- Filled 1 missing values in 'Units_Sold' using median.
- Filled 1 missing values in 'Sales' using median.
- Filled 1 missing values in 'Profit' using median.
- Filled 1 missing values in 'Discount' using median.
- Filled 1 missing values in 'Customer_Segment' using mode.
- Removed unnecessary spaces from text values.

4. Column Information

Column Name	Data Type	Missing Values	Unique Values
Order_ID	int64	0	106
Order_Date	datetime64[us]	0	106
Region	str	0	4
Product	str	0	10
Category	str	0	3
Customer_Segment	str	0	3

Column Name	Data Type	Missing Values	Unique Values
Units_Sold	float64	0	21
Sales	float64	0	26
Profit	float64	0	59
Discount	float64	0	4

5. Machine Learning Results

No machine learning model has been executed for this report.

6. Interpretation and Limitations

Machine learning results depend on the dataset, selected features, data quality, and train-test split. Evaluation scores should not be treated as guarantees of future performance. Correlation does not establish causation. Clustering results describe groupings in the provided data and may not represent naturally distinct populations.

7. Data Privacy

NOVA is designed around local dataset processing. This report contains dataset summaries and analysis results rather than a complete copy of the uploaded dataset.